Credit Card Financial & Customer Analytics

DASHBOARD

Weekly Transaction & Customer Analysis | PostgreSQL · Power BI · DAX

$55M	$7.8M	$576M	3.19/5
Total Revenue	Total Interest	Total Income	Avg Satisfaction

Data Analyst Portfolio Project · GitHub Ready

PROJECT OVERVIEW

🎯 Objective

Build a full-stack Credit Card Analytics solution to help business stakeholders monitor revenue, customer behaviour, and transaction patterns on a weekly basis.

Two Power BI dashboards were created:

- Credit Card Transaction Report – card & spend-level metrics
- Credit Card Customer Report – demographic & income-level metrics

This project demonstrates end-to-end data analyst skills: SQL ingestion → data modelling → DAX engineering → visual storytelling.

⚙ Tech Stack

PostgreSQL

Raw CSV ingestion via COPY command; schema design with 2 normalized tables

Power BI

Two interactive dashboards with slicers, KPI cards & charts

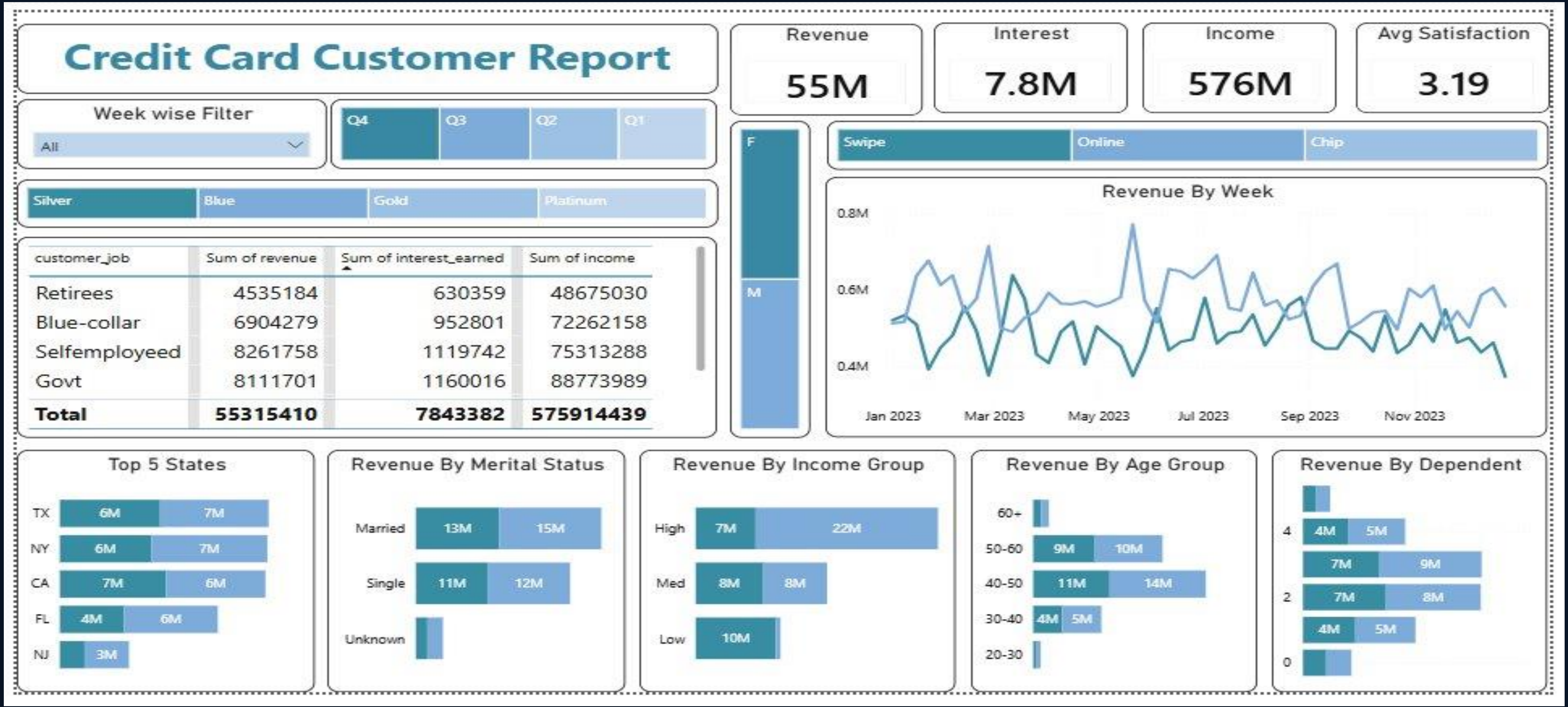
DAX

Revenue formula, Age Group & Income Group calculated columns; WEEKNUM measure

CSV Files

cc_add.csv (transactions) and cust_add.csv (customer demographics)

DASHBOARD 1 — Credit Card Customer Report (All Customers)



Source: Power BI · Credit Card Customer Report · Full dataset (All weeks, All genders)

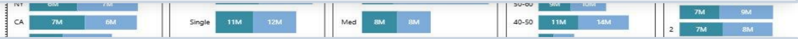
CUSTOMER DASHBOARD — KEY INSIGHTS

💰 Revenue Dominated by Blue-Collar & Govt

Blue-collar ($6.9M) and Govt ($8.1M) job categories drive the most revenue. Despite lower income brackets, high transaction frequency makes these segments the backbone of card revenue.

📍 TX, NY, CA Lead with ~$6-7M Each

The top 3 states each generate $6-7M in revenue. Geographic concentration suggests opportunity for targeted marketing and premium card upsell in these regions.



💍 Married Customers Generate More Revenue

Married segment: $13M vs Singles: $11M. Higher household spending needs and joint financial obligations push married customers toward higher card utilization.

📊 40-50 Age Group is the Sweet Spot

The 40-50 age cohort ($11M female + $14M male) outperforms all others. This group balances high income, stable employment, and established spending habits — ideal for premium card offers.

👨‍👩‍👧‍👦 2-Dependent Households Spend the Most

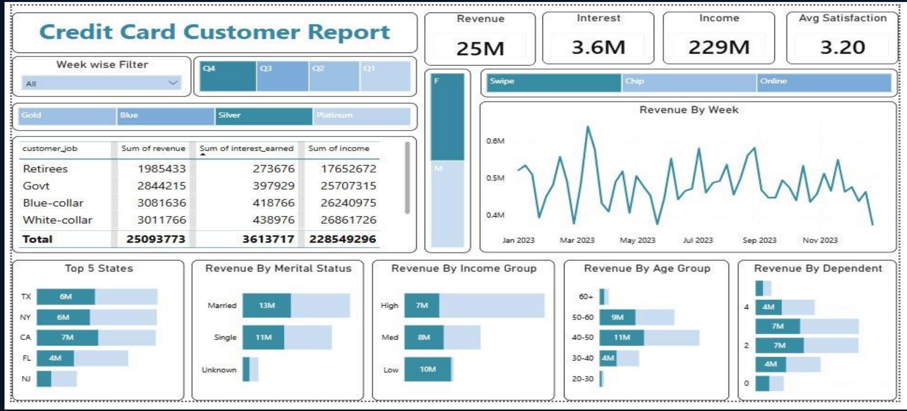
Customers with 2 dependents generate the highest revenue tier. Family financial obligations increase card usage for groceries, fuel, education, and EMI payments.

😊 Satisfaction Score Stable at 3.19-3.20

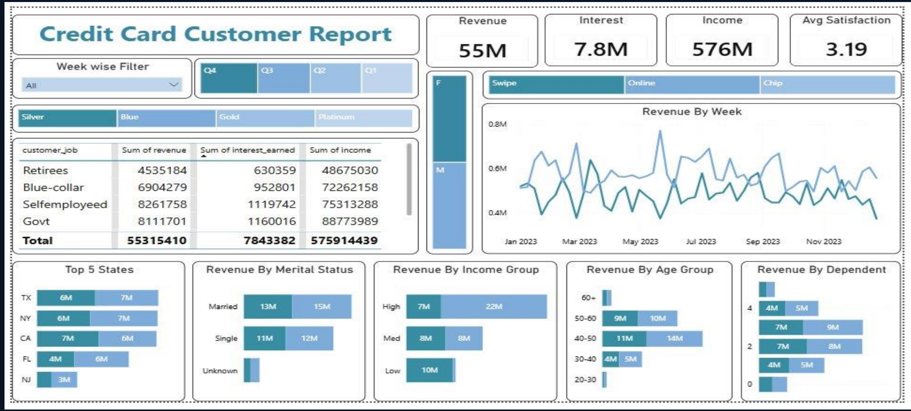
Near-identical satisfaction across gender segments (3.19 vs 3.20) suggests service parity — positive for retention, but headroom exists to push score above 4.0 through targeted benefits.

DASHBOARD 1 — Gender Comparison: Female vs Male Customer Segments

Female Segment

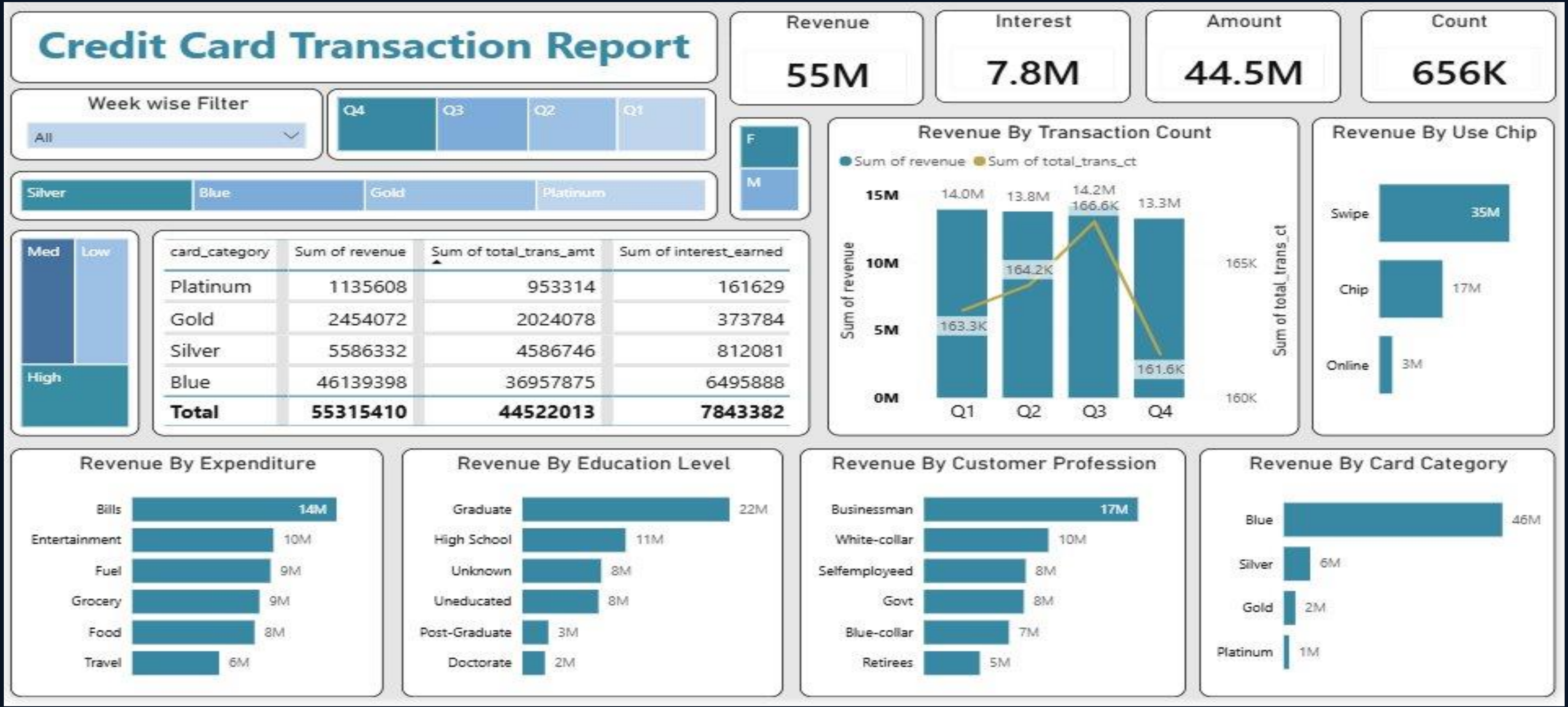


Male Segment



METRIC	FEMALE	MALE
Revenue	$25M	$55M
Interest Earned	$3.6M	$7.8M
Total Income	$229M	$576M
Avg Satisfaction	3.20	3.19

DASHBOARD 2 — Credit Card Transaction Report (Full Dataset · All Cards)



card_category	Sum of revenue	Sum of total_trans_amt	Sum of interest_earned
Platinum	1135608	953314	161629
Gold	2454072	2024078	373784
Silver	5586332	4586746	812081
Blue	46139398	36957875	6495888
Total	**55315410**	**44522013**	**7843382**

Source: Power BI · Credit Card Transaction Report · Full dataset (All card categories, All weeks)

TRANSACTION DASHBOARD — KEY INSIGHTS

💳 Blue Card Dominates at $46M Revenue

Blue card holders generate 84% of total revenue ($46M of $55M). This entry-level card has the widest user base, making it the single most important product segment to monitor and grow.

💼 Businessmen Lead by Profession ($17M)

Business owners generate $17M in revenue — almost double White-collar ($10M). High-spend, low-risk profile makes this segment ideal for premium card migration and rewards upsell.

📚 Graduate Customers Spend the Most ($22M)

Graduates contribute $22M, nearly double High School ($11M). Education level strongly predicts credit capacity and spending sophistication. Targeting post-grads for premium tiers makes strategic sense.

🏷️ Bills & Food are Top Spend Categories

Bills ($14M) and Food ($8M) are non-discretionary categories that dominate spend. These are 'sticky' expenditures that drive consistent transaction activity and revolving balance.

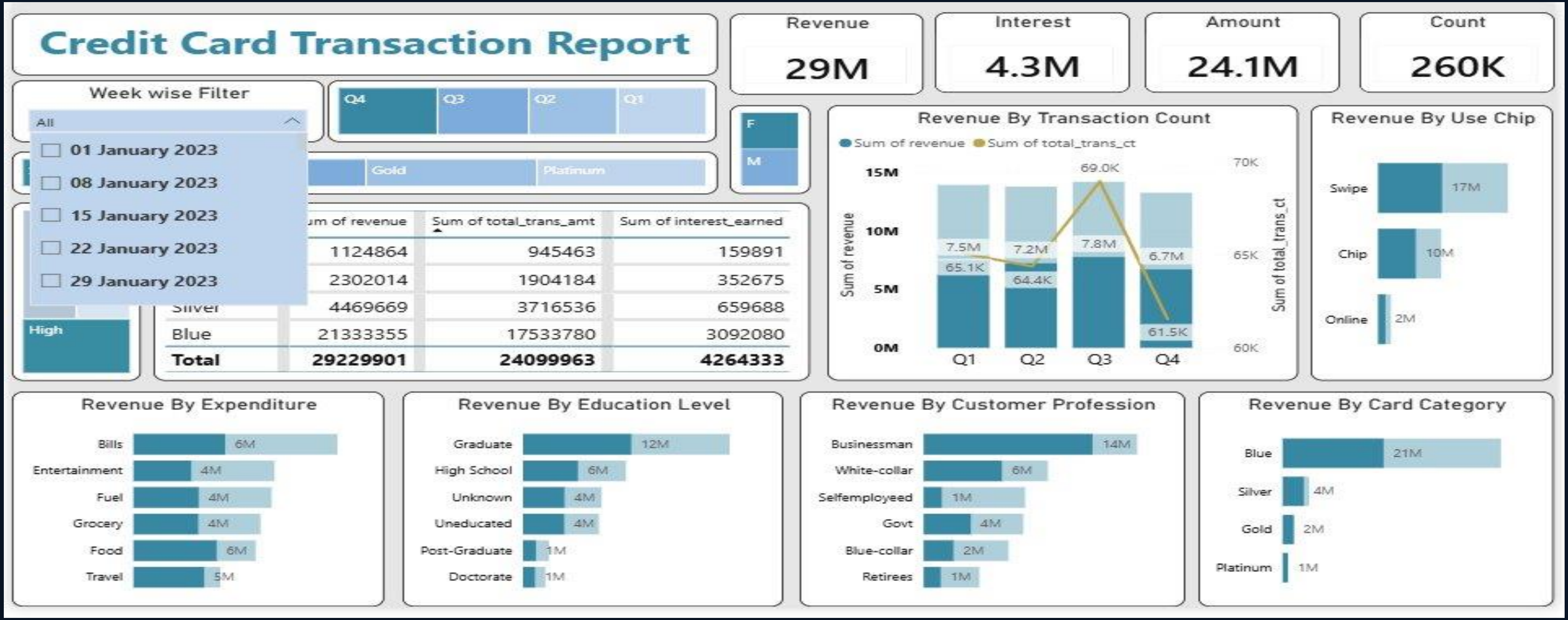
📱 Swipe Transactions Drive 64% of Revenue

Swipe ($35M) massively outperforms Chip ($17M) and Online ($3M). Low online adoption suggests an opportunity to push digital channels with cashback incentives.

📈 Q3 Peak: Highest Revenue & Transaction Count

Q3 shows the highest revenue ($14.2M) and transaction count (166.6K). Summer spending on travel, entertainment, and back-to-school drives the seasonal spike — timing campaigns to Q3 maximizes ROI.

DASHBOARD 2 — Weekly Drill-Down Filter in Action (Jan 2023 Sample)



Weeks Available

01 Jan – 29 Jan 2023
(weekly slicing)

Jan Revenue Snapshot

$29M total
$4.3M interest

Jan Transaction Count

260K transactions
$24.1M amount

DATA PIPELINE & SCHEMA DESIGN

01	02	03	04	05
CSV Files	**PostgreSQL**	**Power BI Connect**	**DAX Measures**	**Dashboard**
cc_add.csv cust_add.csv	CREATE TABLE COPY command	Native PG connector	Revenue, Groups WEEKNUM	2 Reports Published

📋 TABLE 1: cc_details (Transactions)

- client_num (PK/FK)
- card_category
- annual_fees
- activation_30_days
- customer_acq_cost
- credit_limit
- total_revolving_bal
- total_trans_amt
- total_trans_ct
- avg_utilization_ratio
- interest_earned
- delinquent_acc
- use_chip, exp_type

📋 TABLE 2: cc_customers (Demographics)

- Client_Num (PK/JOIN)
- Customer_Age → age_group
- Gender, Marital_Status
- Education_Level
- Customer_Job
- Income → income_group
- Dependent_Count
- state_cd, Zipcode
- Car_Owner, House_Owner
- Personal_loan
- contact
- Cust_Satisfaction_Score

DAX TRANSFORMATIONS (Power BI Calculated Columns & Measures)

Revenue Measure

```
revenue =
  annual_fees
  + interest_earned
  + total_trans_amt
```

💡 **Business Insight**

Unifies 3 income streams into 1 KPI:
Fee income + Interest income +
Spend volume.
Critical for weekly P&L monitoring.

Age Group Column

```
age_group = SWITCH(TRUE(),
  age < 30, "20-30",
  age < 40, "30-40",
  age < 50, "40-50",
  age < 60, "50-60",
  "60+")
```

💡 **Business Insight**

Buckets 10,000+ customers into 5
cohorts.
Enables cohort analysis and
targeted
product/limit strategy per age band.

Income Group Column

```
income_group = SWITCH(
  TRUE(),
  income < 35000,  "Low",
  income < 70000,  "Med",
  income >= 70000, "High"
)
```

💡 **Business Insight**

Three-tier segmentation for credit
risk assessment and personalised
limit & offer recommendations.

Week Number Measure

```
week_num2 = WEEKNUM(
  cc_details
  [week_start_date]
  .[Date]
)
```

💡 **Business Insight**

Maps weekly dates to ISO week
numbers.
Powers the Week-wise filter slicer
for
granular trend tracking across 52
weeks.

END-TO-END ANALYST WORKFLOW

01

Requirements & Data Sourcing

Identified business need: stakeholders needed weekly visibility into credit card revenue, customer demographics, and spend patterns. Sourced raw CSVs: cc_add.csv (transaction-level) and cust_add.csv (customer-level).

02

PostgreSQL Schema Design & Ingestion

Designed 2 normalized tables (cc_details & cc_customers) with correct data types. Used PostgreSQL's COPY command for bulk ingestion. Validated data integrity with SELECT * queries. Tables joined via client_num.

03

Power BI Connection & Data Modelling

Connected Power BI Desktop to PostgreSQL via native connector. Created table relationships. Defined data model structure to support cross-table filtering across card categories, customer demographics, and time.

04

DAX Engineering

Built 4 calculated columns/measures: Revenue (3-stream aggregation), Age Group (5 cohorts via SWITCH), Income Group (3 tiers), and Week Number (WEEKNUM for the time-slicer). Validated outputs against source data.

05

Dashboard Design & Visualisation

Designed 2 dashboards: (1) Transaction Report — card-level, spend-category, and profession-level views; (2) Customer Report — demographic, geographic, and satisfaction views. Both feature Week-wise & Quarter slicers for interactivity.

STRATEGIC RECOMMENDATIONS FOR STAKEHOLDERS

Target 40-50 Age Cohort

Highest-revenue age band. Offer premium Blue→Gold/Platinum upgrade campaigns with reward multipliers on Bills and Travel.

Upsell to Businessmen

$17M revenue already. This segment has capacity for higher limits and Platinum card benefits — lowest churn risk.

Geographic Deep-Dive

TX, NY, CA already dominant. Investigate NJ ($3M) underperformance — is it market size or product-market fit?

Push Online Transactions

Online contributes only $3M vs Swipe $35M. Launch cashback incentives for online spend to improve digital adoption.

Monitor Delinquency Closely

Low-income customers with high utilization ratios (>70%) are most at risk. Weekly delinquent_acc flag monitoring can reduce write-offs.

Q3 Campaign Planning

Q3 is the annual revenue peak. Align marketing spend, credit limit reviews, and product launches to maximise seasonal uplift.

Thank You

Credit Card Financial & Customer Analytics Dashboard · PostgreSQL · Power BI · DAX

🐙 **GitHub:** https://github.com/rajatkumar-analytics/Credit-Card-Financial-Customer-Analytics-Dashboard